

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	MAS-TBVU
Program name	Master of Tall Buildings and Vertical Urbanism / MAS-TBVU
year	2025
Program assessment coordinator	Antony Wood; Yohan Kim
Assessment plan year	2024-25
No assessment this year per plan	☐
LO(s) assessed	LO #2: Students will be able to implement the use of appropriate technologies and systems in approaches to conceptual and technical high-rise building / urban design, while investigating the sustainability implications of high-rise buildings, urban density, and vertical cities.
Presented at program authority meeting date: <i>Please attach minutes</i>	Assessment Faculty: Antony Wood; Yohan Kim

Methodology

LO #2 has been assessed based on the final outputs from **ARCH 546 Architectural Studio VI: Advanced** and **ARCH 573 Tall Building Technologies II**, in accordance with the assessment plan submitted in 2023. These outputs include presentation boards and digital presentations, which are appropriate for evaluating how effectively these courses enable students to achieve the learning objective.

The submitted materials represent the design of sustainable high-rise buildings and their impact on the urban context in ARCH 546, as well as the integration of computational simulation results and technical aspects into the design process in ARCH 573. Additionally, the quality of student work and their efforts are reflected in the submitted materials.

A 5-point grading scale (A, B, C, D, and E) has been used to evaluate student work and determine how well students have met the learning outcomes in each course. Instructors for both ARCH 546 and ARCH 573 assigned letter grades based on the evaluation of final outputs as well as the students' progress throughout the semester.

Please see the ARCH 573 *Tall Building Technologies II* evaluation method below:

- Attendance / Participation in class = 20% tracking each week
- Progress / Final Project & Submission = 80% submission due 05/08/2025

Results

See artifact samples attached.

ARCH 546 Arch Studio VI: Advanced: 100% of students achieved a “B” or higher (5-point scale).

- Proficient (P): Attached, “ARCH546_Sample_P”
- Meets expectations (M): Attached, “ARCH546_Sample_M”
- Does not meet expectations (D): All students meet expectations.

ARCH 573 Tall Building Technologies: 100% of students achieved a “B” or higher (5-point scale).

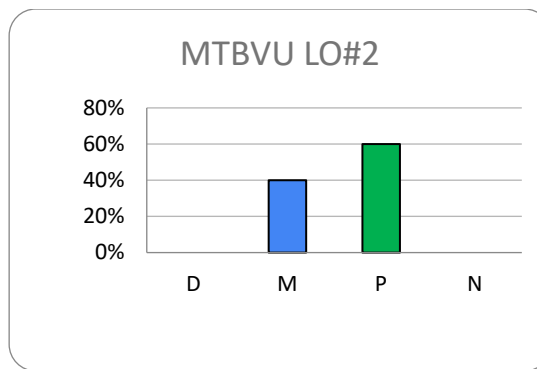
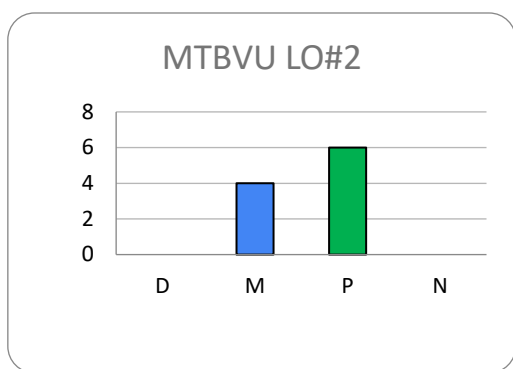
- Proficient (P): Attached, “ARCH573_Sample_P”
- Meets expectations (M): Attached, “ARCH573_Sample_M”
- Does not meet expectations (D): All students meet expectations.

Data table

ARCH 546 Arch Studio VI: Advanced

Class	ARCH546	D	0	0%	does not meet
AY	2025	M	5	38%	meets
program	MTBVU	P	8	62%	proficient
LO#	2	N	0	0%	not assessed
assessor	Wood & Kim	13			
rubric	See the report	TRUE			
# assessed:	13	all			

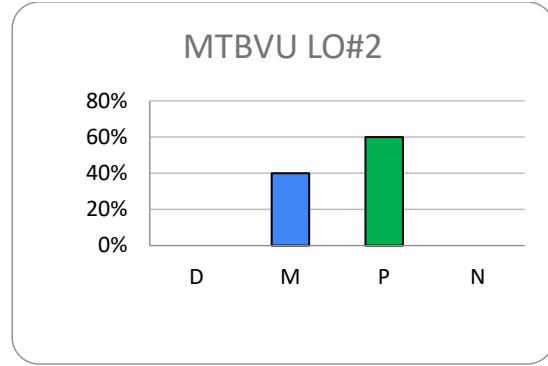
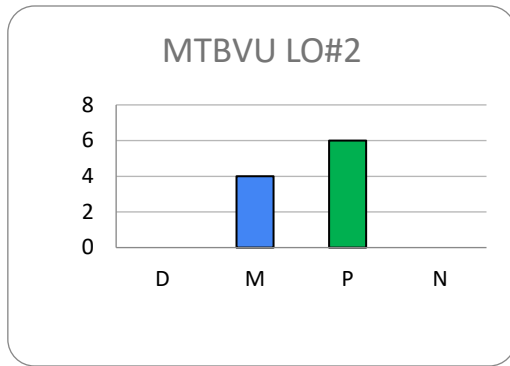
MTBVU LO#2		
student	major	D/M/P
student 1	ARCH	P
student 2	ARCH	P
student 3	ARCH	M
student 4	ARCH	M
student 5	ARCH	M
student 6	ARCH	M
student 7	ARCH	P
student 8	ARCH	P
student 9	ARCH	P
student 10	ARCH	P
student 11	ARCH	M
student 12	ARCH	P
student 13	ARCH	P



ARCH 573 Tall Building Technologies II

Class	ARCH573	D	0	0%	does not meet
AY	2025	M	4	40%	meets
program	MTBVU	P	6	60%	proficient
LO#	2	N	0	0%	not assessed
assessor	Wood & Kim	10			
rubric	See the report	TRUE			
# assessed:	10	all			

MTBVU LO#2		
student	major	D/M/P
student 1	ARCH	M
student 2	ARCH	M
student 3	ARCH	P
student 4	ARCH	P
student 5	ARCH	P
student 6	ARCH	P
student 7	ARCH	P
student 8	ARCH	P
student 9	ARCH	M
student 10	ARCH	M



Description of findings

Most students met expectations and achieved a grade of “B” or higher (on the 5-point scale) in the two courses where LO #2 was assessed. Their final outcomes demonstrate an understanding of the learning objectives and instructions, as well as a strong investment in the work. In fact, more than 50% of the students in each course exceeded expectations.

In ARCH 573 *Tall Building Technologies II*, some students found the computational simulation component challenging due to limited prior experience and understanding. Although they ultimately met expectations or the minimum requirements, this was their first time using simulation software. Many were unfamiliar with concepts such as analyzing airflow behavior and thermal conditions in buildings and integrating these analyses into the design process. While the final projects were successful, students required more time than the instructor anticipated to fully understand the software and its technical details. In addition, it would be ideal for the computational simulation results from ARCH 573 to be more fully integrated into the design projects in ARCH 546. Continuous Improvement (CI) plans to address these challenges are described in the next section.

Continuous Improvement, CI, plans

Prior CI plan impact

Previous Continuous Improvement (CI) plans proposed preparing students in ARCH 570 *Talking Tall I* by introducing research topics early on, followed by conducting in-depth research projects in ARCH 571 *Talking Tall II*. This sequence was intended to help students develop preliminary ideas and reduce difficulties with research, particularly for those less familiar with the process. Additionally, the CI plans suggested exploring potential overlaps between ARCH 571 and ARCH 546 *Architectural Studio VI: Advanced* to investigate how research findings could inform design decisions. However, these plans had a limited impact during this academic year due to a shift in ARCH 571 from a research-oriented format to a traditional lecture/seminar course.

This cycle's CI plan

A proposed improvement would be for ARCH 573 *Talking Tall II* instructor to convene a few additional sessions (e.g., workshops) outside regular class time at the beginning of the semester, focused on the computational simulation software that students are expected to use throughout the course. It would also be beneficial to encourage students to take advantage of office hours, in addition to these sessions, to further support their learning of the software.

Since ARCH 573 is offered in the spring semester, the instructor could provide students with online tutorials to explore during the winter break, helping them become familiar with the software in advance. This approach could reduce the number of in-class sessions devoted to basic simulation tutorials, allowing more time for in-depth discussions on post-processing, simulation results, and their implications for studio design projects. As ARCH 573 is open only to students enrolled in the MTBVU program, it would not be difficult to coordinate with students ahead of the spring semester.

Additionally, there is potential for greater integration between ARCH 573 and ARCH 546. Specifically, students could be asked to incorporate simulation results from ARCH 573 into their studio projects from the beginning of the semester, positioning the course as a supporting or companion course to the design studio. For example, weekly assignments in ARCH 573 could align with the design progress in ARCH 546. This would allow students to explore how computational simulation—particularly airflow behavior and thermal conditions—can inform and enhance design decisions, especially in the context of tall buildings. These plans could be adapted based on the specific topics covered in future ARCH 546 studios.

Lastly, the faculty review committee, in consultation with professors teaching relevant courses, will make appropriate recommendations to the faculty and the Dean for changes in courses. Student evaluation results will be reviewed to enhance the courses and bridge the gap between course objectives and student expectations.